

EBS Volume type Notes

1. EBS Volume Types

Amazon Elastic Block Store volumes are divided into **two categories**:

1. **SSD Based Volumes**
2. **HDD Based Volumes**

Total **6 types** of EBS volumes.

2. SSD Based Volumes (Fast Storage)

SSD volumes provide **high performance and low latency**.

1. gp3 (General Purpose SSD)

- Most commonly used volume
- Balanced **price and performance**
- Used for web servers and applications

2. gp2 (General Purpose SSD)

- Older version of gp3
- IOPS automatically increases with size

3. io1 / io2 (Provisioned IOPS SSD)

- Very high performance
- Used for **databases**

Example:

- MySQL
 - PostgreSQL
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3. HDD Based Volumes (Cheap Storage)

HDD volumes are used for **large data storage**.

4. st1 (Throughput Optimized HDD)

Used for:

- Big data
- Log processing
- Data warehouse

5. sc1 (Cold HDD)

- Very cheap storage
- Used for **backup and rarely used data**

6. Magnetic (Standard)

- Old generation storage
 - Rarely used today
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4. Simple Table

Type	Category	Use
gp3	SSD	Most common
gp2	SSD	Old general purpose
io1 / io2	SSD	High performance database
st1	HDD	Big data
sc1	HDD	Cold storage
Magnetic	HDD	Old type

5. SSD and HDD Full Form

SSD

SSD = Solid State Drive

- Very fast storage
- No moving parts
- Data stored in flash memory

Uses:

- Operating system
 - Databases
 - Cloud servers like Amazon EC2
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HDD

HDD = Hard Disk Drive

- Old storage technology
 - Has spinning disk
 - Uses mechanical arm to read and write data
 - Slower but cheaper than SSD
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6. SSD vs HDD

Feature	SSD	HDD
Speed	Fast	Slow
Moving Parts	No	Yes
Price	Expensive	Cheap

Simple Example

SSD → Mobile phone storage
HDD → Old computer hard disk

Important Point for Revision

Most DevOps engineers use:

gp3 SSD volume